Quadratic Equation

Tue May 05 2026



Bhaskara Sarma VNV

31 Years of Experience

QUADRATIC EQUATION

$a \neq 0$ ℝ & ℂ

Equation of the form $ax^2 + bx + c = 0$ where $a, b, c \in R$ and $a \neq 0$ is called a quadratic equation, where a, b, c are coefficients of this equation.

Quadratic equation has exactly two roots. It can be real or complex in nature and roots can be distinct or repetitive.

≤ 2

$ax^2+bx+c=0$
$a(x-\alpha)(x-\beta)=0$
$x=\alpha,\beta$

Definition Root of a quadratic equation

$\alpha \in \mathbb{R}$ or $\mathbb{C}$ $\quad a\alpha^2 + b\alpha + c = 0$

Number of roots of a quadratic equation

$\leq$ degree ≤ 2

- 0 real roots
- 1 real root
- 2 real roots

How to find roots of a quadratic equation

- factorisation
- formula

Standard Quadratic Equation

An equation of the form

$$ax^2 + bx + c = 0 \quad \ldots(i)$$

where $a, b, c \in C$ and $a \neq 0$, is called a standard quadratic equation.

The numbers a, b, c are called the coefficients of this equation.

Definition: Root of a Quadratic Equation

A root of the quadratic Eq. (i) is a complex number α, such that $a\alpha^2 + b\alpha + c = 0$. Recall that $D = b^2 - 4ac$ is the discriminant of the Eq. (i) and its roots are given by the following formula.

$$x = \frac{-b \pm \sqrt{D}}{2a} \quad \text{[Shridharacharya method]}$$

Roots of Quadratic Equation

To find roots of quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$, first divide it by a.

we get $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

$$\Rightarrow \quad \left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0$$

$$\Rightarrow \quad \left(x + \frac{b}{2a}\right)^2 = \left(\frac{b^2 - 4ac}{4a^2}\right)$$

$$\Rightarrow \quad x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$\Rightarrow \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$ these are the roots of the given equation.

$$ax^2 + bx + c \equiv a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a}$$

$$ax^2 + bx + c \equiv a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a}$$

Relation Between Roots and Coeffiecients

Let the roots are α and β then $\alpha = \dfrac{-b-\sqrt{b^2-4ac}}{2a}$

$\beta = \dfrac{-b+\sqrt{b^2-4ac}}{2a}$

$a \neq 0$

(a) sum of the roots $= \alpha+\beta = -\dfrac{b}{a}$

$S_1 = \alpha+\beta = -\dfrac{b}{a}$

(b) product of the roots $= a\beta = \dfrac{c}{a}$

$S_2 = \alpha\beta = \dfrac{c}{a}$

(c) If α and β are the roots of $ax^2+bx+c=0$ then, $ax^2+bx+c = a(x-\alpha)(x-\beta)$

(d) The equation whose roots are α and β is

$a(x-\alpha)(x-\beta)=0$

$\Rightarrow x^2-(\alpha+\beta)x+a\beta=0$

$\Rightarrow x^2-(S_1)x+(S_2)=0$

$\Rightarrow x^2-(\text{sum of roots})x+(\text{product of roots})=0$

$x^2 - S_1 x + S_2 = 0$

$a(x-\alpha)(x-\beta)=0$

$a(x^2-(\text{sum of roots})x+\text{product of roots})=0$

S_1 S_2

$\alpha+\beta=-b/a$ & $\alpha\beta=c/a$

$a\alpha^2+b\alpha+c=0$

$a\beta^2+b\beta+c=0$

$ax^2+bx+c=a(x-\alpha)(x-\beta)$

$ax^2+bx+c=0$ has roots α and β

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$S_1 = \alpha+\beta=0$

$S_2 = \alpha\beta=0$

Relation between Roots and Coefficients

If roots of quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ are and β then

(i) $|\alpha - \beta| = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \left|\frac{\sqrt{b^2-4ac}}{a}\right| = \left|\frac{\sqrt{D}}{a}\right|$

(ii) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = \frac{b^2-2ac}{a^2}$

(iii) $|\alpha^2 - \beta^2| = |\alpha+\beta|\sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \left|\frac{b\sqrt{b^2-4ac}}{a^2}\right|$

(iv) $\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = -\frac{b(b^2-3ac)}{a^3}$

Relation between Roots and Coefficients

(v) $\alpha^3 - \beta^3 = (\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta)$

$= (\alpha - \beta)\{(\alpha + \beta)^2 - \alpha\beta\}$

(vi) $\alpha^4 + \beta^4 = \{(\alpha + \beta)^2 - 2\alpha\beta\}^2 - 2\alpha^2\beta^2$

(vii) $\alpha^4 - \beta^4 = (\alpha^2 - \beta^2)(\alpha^2 + \beta^2)$

$= (\alpha - \beta)(\alpha + \beta)((\alpha + \beta)^2 - 2\alpha\beta)$

(viii) $\alpha^2 + \alpha\beta + \beta^2 = (\alpha + \beta)^2 - \alpha\beta$

(ix) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$

(x) $\alpha^2\beta + \beta^2\alpha = \alpha\beta(\alpha + \beta)$

(xi) $\left(\frac{\alpha}{\beta}\right)^2 + \left(\frac{\beta}{\alpha}\right)^2 = \frac{\alpha^4 + \beta^4}{\alpha^2\beta^2} = \frac{(\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2}{\alpha^2\beta^2}$

Formation of an Equation with Given Roots

A quadratic equation whose roots are α and β is given by $(x - \alpha)(x - \beta) = 0$

$\therefore x^2 - \alpha x - \beta x + \alpha\beta = 0 \Rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$

In other words required equation is x^2 − (sum of roots)x + product of roots = 0

2. Formation of an equation with given roots

A quadratic equation whose roots are α and β, is given by $(x - \alpha)(x - \beta) = 0$ or $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

i.e. x^2 − (Sum of roots) x + Product of roots $= 0$

$\therefore \quad x^2 - Sx + P = 0.$

$S_1 = \alpha + \beta$

$P = \alpha\beta$

$x^2 - S_1 x + S_2 = 0$

Relation between Roots and Coefficients

1. **Relation between roots and coefficients of quadratic equation** If roots of the equation $ax^2 + bx + c = 0\ (a \neq 0)$ be real and distinct and $\alpha < \beta$,

then $\alpha = \dfrac{-b + \sqrt{D}}{2a}, \beta = \dfrac{-b - \sqrt{D}}{2a}$.

(i) Sum of roots

$$= S = \alpha + \beta = -\frac{b}{a} = -\frac{\text{Coefficient of } x}{\text{Coefficient of } x^2}.$$

(ii) Product of roots

$$= P = \alpha\beta = \frac{c}{a} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}.$$

(iii) Difference of roots

$$= D' = \alpha - \beta = \frac{\sqrt{D}}{a} = \frac{\sqrt{\text{Discriminant}}}{\text{Coefficient of } x^2}.$$

$= \dfrac{\sqrt{D}}{|a|}$

$(a-b)^2 =$

$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$

$|\alpha-\beta| = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$

$|\alpha-\beta| = \sqrt{S_1^2 - 4S_2}$

3. Symmetric function of roots A function of α and β is said to be symmetric function, if it remains unchanged, when α and β are interchanged.

For example, $\alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$ is a symmetric function of α and β.

In order to find the value of a symmetric function in terms of $\alpha + \beta$, $\alpha\beta$ and $\alpha - \beta$ and also in terms of a, b and c.

(i) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) = \frac{b^2 - 2ac}{a^2}.$$

(ii) $\alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$

$$= \left(-\frac{b}{a}\right)\left(\frac{\sqrt{D}}{a}\right) = -\frac{b\sqrt{D}}{a^2}.$$

(iii) $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

$$=\left(-\frac{b}{a}\right)^3-3\left(\frac{c}{a}\right)\left(-\frac{b}{a}\right)=-\left(\frac{b^3-3abc}{a^3}\right).$$

(iv) $\alpha^3-\beta^3=(\alpha-\beta)^3+3\alpha\beta(\alpha-\beta)$

$$=\left(\frac{\sqrt{D}}{a}\right)^3+3\left(\frac{c}{a}\right)\left(\frac{\sqrt{D}}{a}\right)=\frac{\sqrt{D}(D+3ac)}{a^3}.$$

(v) $\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2$

$$=\left(\frac{b^2-2ac}{a^2}\right)^2-2\left(\frac{c}{a}\right)^2=\frac{b^4+2a^2c^2-4acb^2}{a^4}.$$

(vi) $\alpha^4-\beta^4=(\alpha^2+\beta^2)(\alpha^2-\beta^2)$ $(\alpha-\beta)(\alpha+\beta)$

$$=-\frac{b\sqrt{D}(b^2-2ac)}{a^4}.$$

(vii) $\alpha^5+\beta^5=(\alpha^2+\beta^2)(\alpha^3+\beta^3)-\alpha^2\beta^2(\alpha+\beta)$

Note!

$$=\left(\frac{b^2-2ac}{a^2}\right)\left(-\frac{(b^3-3abc)}{a^3}\right)-\frac{c^2}{a^2}\left(-\frac{b}{a}\right)$$

$$=\frac{-(b^5-5ab^3c+5a^2bc^2)}{a^5}.$$

(viii) $\alpha^5-\beta^5=(\alpha^2+\beta^2)(\alpha^3-\beta^3)+\alpha^2\beta^2(\alpha-\beta)$

$$=\left(\frac{b^2-2ac}{a^2}\right)\left(\frac{\sqrt{D}\,(D+3ac)}{a^3}\right)+\left(\frac{c}{a}\right)^2\left(\frac{\sqrt{D}}{a}\right)$$

$$=\frac{\sqrt{D}\,(b^4-3acb^2+3a^2c^2)}{a^5}.$$

Transformation of Quadratic Equations

Let α, β be the roots of the equation $ax^2 + bx + c = 0$, then the equation

(i) whose roots are $\alpha + k, \beta + k$, is

$a(x-k)^2 + b(x-k) + c = 0$ [replace x by $(x-k)$]

(ii) whose roots are $\alpha - k, \beta - k$, is

$a(x+k)^2 + b(x+k) + c = 0$ [replace x by $(x+k)$]

(iii) whose roots are $\alpha k, \beta k$, is

$ax^2 + kbx + k^2 c = 0$ $\left[\text{replace } x \text{ by } \left(\frac{x}{k}\right)\right]$

(iv) whose roots are $\frac{\alpha}{k}, \frac{\beta}{k}$, is

$ak^2x^2 + bkx + c = 0$ [replace x by xk]

(v) whose roots are $-\alpha, -\beta$, is

$ax^2 - bx + c = 0$ [replace x by $(-x)$]

(vi) whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$, is

$cx^2 + bx + a = 0$ $\left[\text{replace } x \text{ by } \left(\frac{1}{x}\right)\right]$

$f(x) \equiv ax^2 + bx + c = 0$

α, β

$f(x-k) = 0$

$a(x-k)^2 + b(x-k) + c = 0$

$f(x+k) = 0$

$f\left(\frac{x}{k}\right) = 0$

$f(kx) = 0$

$f(-x) = 0$

$f\left(\frac{1}{x}\right) = 0$

(vii) whose roots are $-\frac{1}{\alpha}, -\frac{1}{\beta}$, is

$$cx^2 - bx + a = 0$$

[replace x by $\left(-\frac{1}{x}\right)$]

(viii) whose roots are $\frac{k}{\alpha}, \frac{k}{\beta}$, is

$$cx^2 + kbx + k^2a = 0$$

[replace x by $\left(\frac{k}{x}\right)$]

(ix) whose roots are $p\alpha + q, p\beta + q$, is

$$a\left(\frac{x-q}{p}\right)^2 + b\left(\frac{x-q}{p}\right) + c = 0$$

[replace x by $\left(\frac{x-q}{p}\right)$]

(x) whose roots are $\alpha^n, \beta^n, n \in N$, is

$$a(x^{1/n})^2 + b(x^{1/n}) + c = 0$$

[replace x by $(x^{1/n})$]

(xi) whose roots are $\alpha^{1/n}, \beta^{1/n}, n \in N$ is

$$a(x^n)^2 + b(x^n) + c = 0$$

[replace x by (x^n)]

Formation of an Equation with Given Roots

A quadratic equation whose roots are α and β is given by $(x - \alpha)(x - \beta) = 0$

$\therefore \quad x^2 - \alpha x - \beta x + \alpha\beta = 0 \quad \Rightarrow \quad x^2 - (\alpha + \beta)x + \alpha\beta = 0$

In other words required equation is x^2 – (sum of roots)x + product of roots = 0

Equation in terms of the roots of another equation

If α, β are roots of the equation $ax^2 + bx + c = 0$ then the new equation whose roots are

(i) $-\alpha, -\beta \Rightarrow ax^2 - bx + c = 0$ (replace x by $-x$)

(ii) $1/\alpha, 1/\beta \Rightarrow cx^2 + bx + a = 0$ (replace x by $1/x$)

(iii) $\alpha^n, \beta^n; n \in N \Rightarrow a(x^{1/n})^2 + b(x^{1/n}) + c = 0$ (replace x by $x^{1/n}$)

(iv) $k\alpha, k\beta \Rightarrow ax^2 + kbx + k^2c = 0$ (replace x by x/k)

(v) $k + \alpha, k + \beta \Rightarrow a(x - k)^2 + b(x - k) + c = 0$ (replace x by $x - k$)

(vi) $\frac{\alpha}{k}, \frac{\beta}{k} \Rightarrow k^2ax^2 + kbx + c = 0$ (replace x by kx)

(vii) $\alpha^{1/n}, \beta^{1/n}; n \in N \Rightarrow a(x^n)^2 + b(x^n) + c = 0$ (replace x by x^n)

Note:

(i) This method can also be applied to higher degree polynomial equations.

(ii) Above method is valid when same transformation is applied to both new roots

NATURE OF ROOTS

Let discuss roots of quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in R$,

As we know roots are given by $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

To understand the nature of roots we need to analyze the term written in square root as $b^2 - 4ac = D$ (let Discriminant),

we can rewrite roots as $x = \dfrac{-b \pm \sqrt{D}}{2a}$

$D = b^2 - 4ac$

Discriminant ($D = b^2 - 4ac$)	Nature of Roots
$D > 0$	Real and Distinct ($\alpha, \beta \in R$ and $\alpha \neq \beta$)
$D = 0$	Real and Equal $(\alpha = \beta = -\frac{b}{2a})$ Quadratic expression can be written as perfect square of linear polynomial $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2$
$D < 0$	Imaginary (non-real) and Distinct ($\alpha, \beta \notin R$ and $\alpha \neq \beta$) The roots are complex conjugates of each other $\alpha = p + iq$ and $\beta = p - iq$ where p and $q \in R$

Nature of Roots

1. If $a, b, c \in R$ and $a \neq 0$, then

(i) If $D < 0$, then Eq. (i) has non-real complex roots.

(ii) If $D > 0$, then Eq. (i) has real and distinct roots, namely

$$x_1 = \frac{-b + \sqrt{D}}{2a}, x_2 = \frac{-b - \sqrt{D}}{2a} \text{ and then}$$

$$ax^2 + bx + c = a(x - x_1)(x - x_2). \quad \ldots(ii)$$

(iii) If $D = 0$, then Eq. (i) has real and equal roots, then

$$x_1 = x_2 = -\frac{b}{2a} \text{ and then}$$

$$ax^2 + bx + c = a(x - x_1)^2. \quad \ldots(iii)$$

To represent the quadratic $ax^2 + bx + c$ in form Eqs. (ii) or (iii), is to expand it into linear factors.

(iv) If $D \geq 0$, then Eq. (i) has real roots.

2. If $a, b, c \in Q$ and D is a perfect square of a rational number, the roots are rational and in case it is not a perfect square, the roots are irrational.

3. If $a, b, c \in R$ and $p + iq$ is one root of Eq. (i) $(q \neq 0)$, then the other must be the conjugate $(p - iq)$ and *vice-versa* (where, $p, q \in R$ and $i = \sqrt{-1}$).

4. If $a, b, c \in Q$ and $p + \sqrt{q}$ is one root of Eq. (i), then the other must be the conjugate $p - \sqrt{q}$ and *vice-versa* (where, p is a rational and $\sqrt{q}$ is a surd).

5. If $a = 1$ and $b, c \in I$ and the roots of Eq. (i) are rational numbers, these roots must be integers.

Sum of coefficients = 0

6. If $a+b+c=0$ and a, b, c are rational, 1 is a root of the Eq. (i) and roots of the Eq. (i) are rational.

$\alpha\beta = \frac{c}{a}$

(1) $\beta = \frac{c}{a}$

$\beta = c/a$

7. $a^2+b^2+c^2-ab-bc-ca=\frac{1}{2}$

$\{(a-b)^2+(b-c)^2+(c-a)^2\}$

$=-\{(a-b)(b-c)+(b-c)(c-a)+(c-a)(a-b)\}$

$x^2-4x+3=0$

1 | 1 −4 3

1 −3 0

$x^2-3x+2=0$

1 −3 2

1 −1 −2 0

Special Cases

Let discuss some special cases of nature of roots of quadratic equation $ax^2 + bx + c = 0$
where $a, b, c \in R$,

(a) If a, b, c are rational numbers and $D = b^2 - 4ac$ is a perfect square of a rational number then the roots are rational.

(b) If a, b, c are rational numbers and $D = b^2 - 4ac$ is not a perfect square then the roots are irrational conjugates of each other. If $p + \sqrt{q}$ is one root then $p - \sqrt{q}$ is other root.

(c) If $a = 1$ and $b, c \in I$ and $D = b^2 - 4ac$ is a perfect square of an integer then quadratic equation has integral roots

(d) If $a + b + c = 0$ then 1 is a root of the equation and if a, b, c are rational numbers then both the roots are rational.

$x^2 - 3x + 2 = 0$

$9 - 4(1)(2) = 1 = 1^2$

$x = 1, 2$

COMMON ROOT(S) OF QUADRATIC EQUATIONS

Consider two quadratic equations,

$$a_1x^2 + b_1x + c_1 = 0 \text{ and } a_2x^2 + b_2x + c_2 = 0$$

For Two Common Roots

- If two quadratic equation has both roots common then two equations should be identical for that, and coefficients of equations are proportional then the ratio of coefficients must be same. hence we get

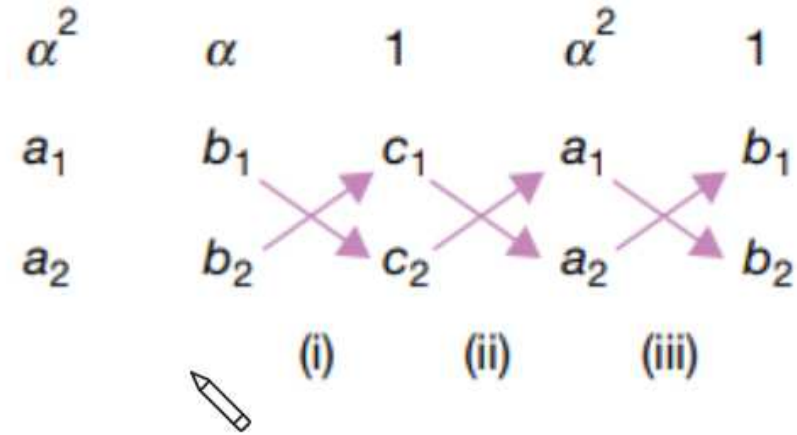
$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

- Two quadratic equations with real coefficients has a non-real root as common root then both roots of given quadratic equations are common as imaginary roots always occur in pair.
- Two quadratic equations with rational coefficients has an irrational root as a common root then both roots of given quadratic equations are common as surd roots always occur in pair.

Let $x = \alpha$ be the common root of two equations. So α should satisfy the two equations.

$$a_1\alpha^2 + b_1\alpha + c_1 = 0 \text{ and } a_2\alpha^2 + b_2\alpha + c_2 = 0$$

Solving the two equations



We get $\dfrac{\alpha^2}{(b_1c_2 - b_2c_1)} = \dfrac{\alpha}{(c_1a_2 - c_2a_1)} = \dfrac{1}{(a_1b_2 - a_2b_1)}$ ...(1)

$$\Rightarrow \quad \alpha^2 = \left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}\right) \text{ and } \alpha = \left(\frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}\right)$$

$$(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)\,(a_1b_2 - a_2b_1)$$

It is a condition for one common root

Note:

To find the common root between the two equations, remove the term containing x^2 term from both the equations and find the value of x, this is the common root.

Condition for Common Roots

1. Only One Root is Common

Consider two quadratic equations

$$ax^2 + bx + c = 0 \text{ and } a'x^2 + b'x + c' = 0$$

[where $a, a' \neq 0$ and $ab' - a'b \neq 0$]

Let α be a common root, then

$$a\alpha^2 + b\alpha + c = 0 \text{ and } \alpha'\alpha^2 + b'\alpha + c' = 0.$$

On solving these two equations by cross-multiplication, we have

$$\frac{\alpha^2}{bc' - b'c} = \frac{\alpha}{ca' - c'a} = \frac{1}{ab' - a'b}$$

From first two relations, we get

$$\alpha = \frac{bc' - b'c}{ca' - c'a} \quad \ldots(i)$$

and from last two relations, we get

$$\alpha = \frac{ca' - c'a}{ab' - a'b} \quad \ldots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{bc' - b'c}{ca' - c'a} = \frac{ca' - c'a}{ab' - a'b}$$

$$\Rightarrow \quad (ab' - a'b)(bc' - b'c) = (ca' - c'a)^2$$

or $$\begin{vmatrix} a_1 & b_1 \\ a'_2 & b'_1 \end{vmatrix} \times \begin{vmatrix} b_1 & c_1 \\ b'_2 & c'_2 \end{vmatrix} = \begin{vmatrix} c_1 & a_1 \\ c'_2 & a'_1 \end{vmatrix}^2$$ [remember]

This is the required condition for one root of two quadratic equations to be common.

2. Both Roots are Common

Let α, β be the common roots of the equations $ax^2 + bx + c = 0$ and $a'x^2 + b'x + c' = 0$, then

$$\alpha + \beta = -\frac{b}{a} = -\frac{b'}{a'} \Rightarrow \frac{a}{a'} = \frac{b}{b'} \quad \ldots(iii)$$

and

$$\alpha\beta = \frac{c}{a} = \frac{c'}{a'} \Rightarrow \frac{a}{a'} = \frac{c}{c'} \quad \ldots(iv)$$

From Eqs. (iii) and (iv), we get $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$

This is the required condition for both roots of two quadratic equations to be identical.

Remark

To find the common root between the two equations, make the same coefficient of x^2 in both equations and then subtract of the two equations.

Quadratic Expression

An expression of the form $ax^2 + bx + c$, where $a, b, c \in R$ and $a \neq 0$ is called a quadratic expression in x. So, in general quadratic expression is represented by

$$f(x) = ax^2 + bx + c \text{ or } y = ax^2 + bx + c.$$

Graph of a Quadratic Expression

We have, $y = ax^2 + bx + c = f(x), \quad [a \neq 0]$

$$\Rightarrow \quad y = a\left[\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a^2}\right]$$

or $$\left(y + \frac{D}{4a}\right) = a\left(x + \frac{b}{2a}\right)^2$$

Now, let $y + \frac{D}{4a} = Y$ and $x + \frac{b}{2a} = X$

$$\therefore \quad Y = aX^2$$

$$\Rightarrow \quad X^2 = \frac{Y}{a}$$

1. The shape of the curve $y = f(x)$ is parabolic.

2. The axis of parabola is $X = 0$ or $x + \frac{b}{2a} = 0$

   or $x = -\frac{b}{2a}$ i.e. parallel to Y-axis.

3. (i) If $\frac{1}{a} > 0 \Rightarrow a > 0$, the parabola open upwards.



(ii) If $\frac{1}{a} < 0 \Rightarrow a < 0$, the parabola open downwards.



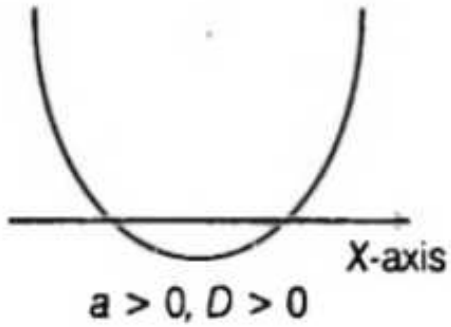
4. Intersection with axes

(i) Intersection with X-axis

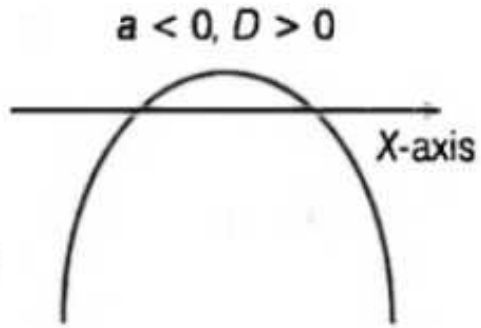
For X-axis, $y = 0$.

$$\therefore \quad ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{D}}{2a}$$

For $D > 0$, parabola cuts X-axis in two real and distinct points



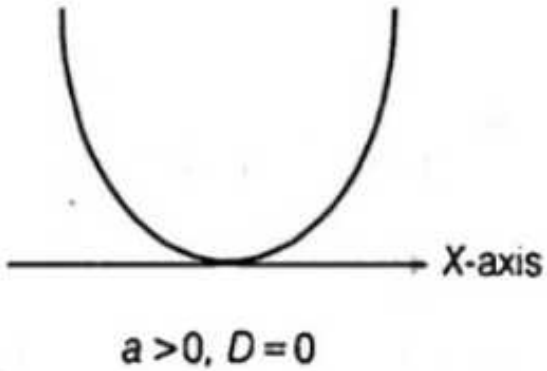
a > 0, D > 0



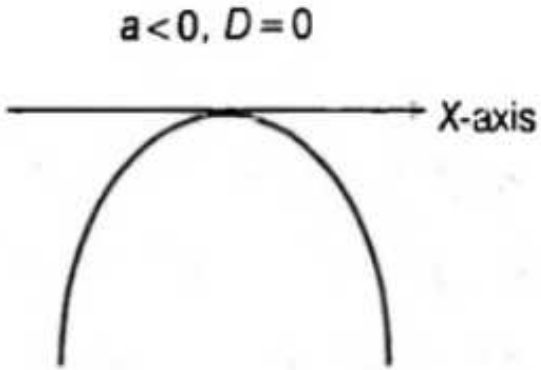
a < 0, D > 0

For $D = 0$, parabola touches X-axis in one point

i.e., $x = -\frac{b}{2a}$.

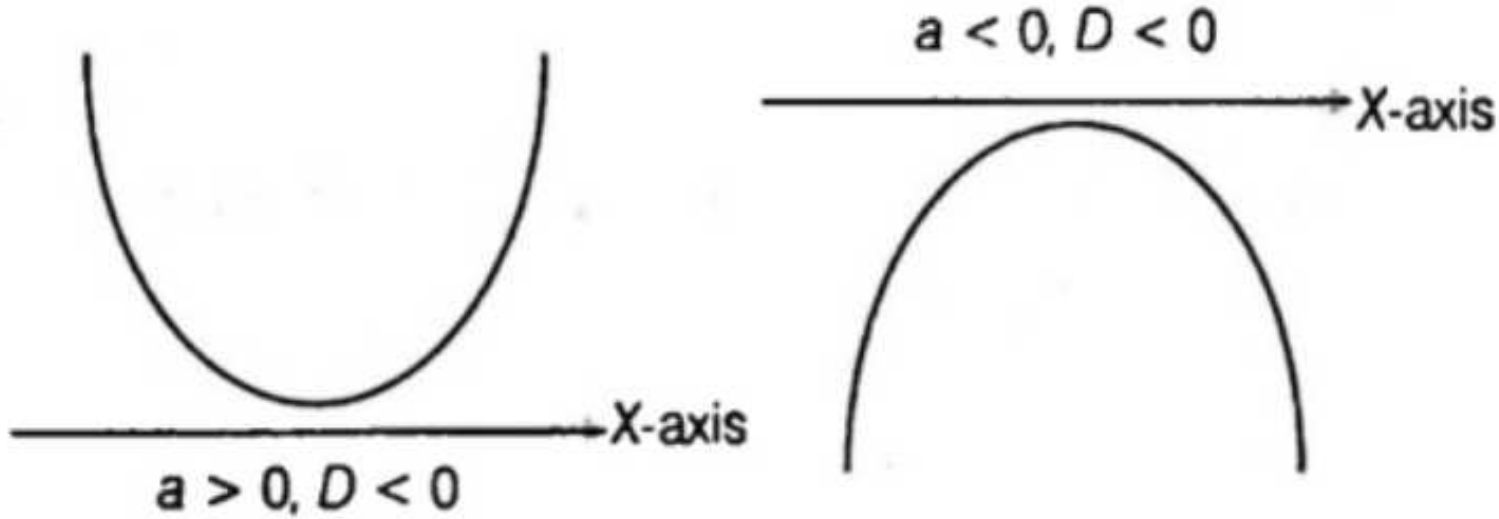


a > 0, D = 0



a < 0, D = 0

For $D < 0$, parabola does not cut X-axis i.e., imaginary values of x.



(ii) Intersection with Y-axis

For Y-axis, $x = 0$.

$\therefore \qquad y = c$

Graph of Quadratic Expression

Let $y = f(x) = Ax^2 + Bx + C$ where $A, B, C \in R$ and $A \neq 0$, four plotting any quadratic expression we just need four steps

(I) Observe the sign of A (coefficient of x^2)
if $A > 0$ (upward Parabola)



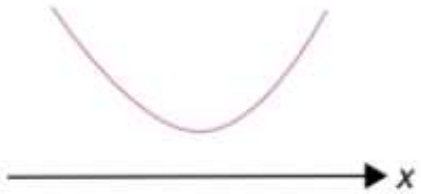
If $A < 0$ (Downward Parabola)



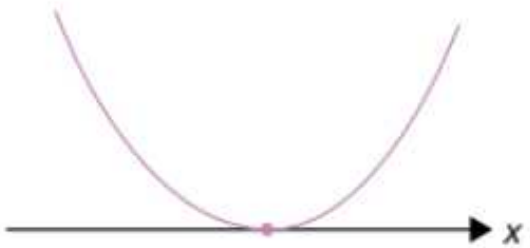
(II) Get discriminant $D = B^2 - 4AC$ then exactly one case can happen among following 6 cases depending upon the value of a and D

CASE-1: $(A > 0)$

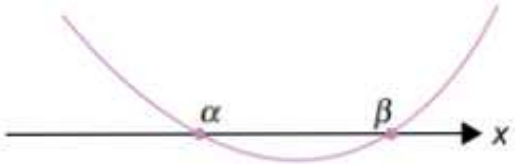
(i) $D < 0$ (Imaginary Roots)



(ii) $D = 0$ (Real and Equal Roots)

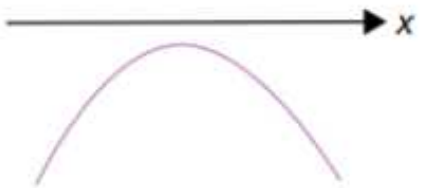


(iii) $D > 0$ (Real and Distinct Roots)

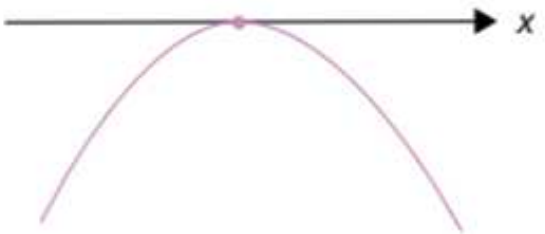


CASE-2: $(A < 0)$

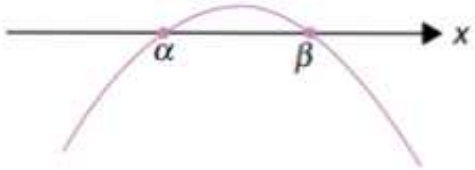
(i) $D < 0$ (Imaginary Roots)



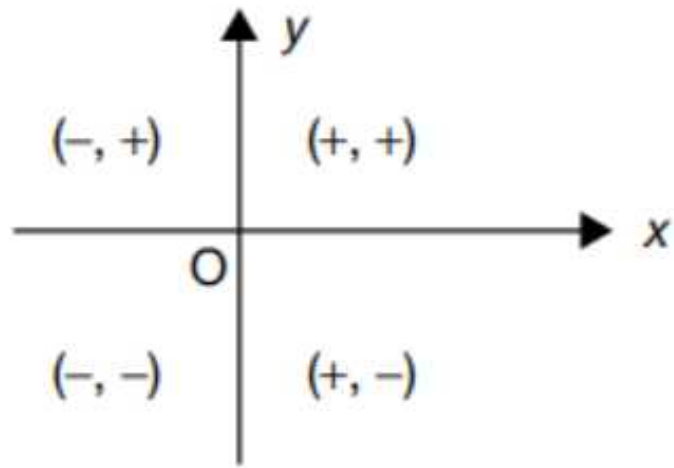
(ii) $D = 0$ (Real and Equal Roots)



(iii) $D > 0$ (Real and Distinct Roots)



(III) Get vertex $V\left(-\frac{B}{2A}, -\frac{D}{4A}\right)$ and locate it in xy plane and identify the quadrant.



(IV) Quadratic expression $y = Ax^2 + Bx + C$ cuts y-axis at point $(0, C)$

GREATEST AND LEAST VALUE OF QUADRATIC EXPRESSION

Expression of the form $ax^2 + bx + c$, where $a, b, c \in R$ and $a \neq 0$ is called a quadratic expression (quadratic polynomial) in x so in general quadratic expression is represented by

$$y = f(x) = ax^2 + bx + c$$

$$= a\left\{\left(x+\frac{b}{2a}\right)^2 + \left(\frac{4ac-b^2}{4a^2}\right)\right\} = a\left(x+\frac{b}{2a}\right)^2 - \frac{D}{4a}$$

where $D = b^2 - 4ac$

$$\Rightarrow \quad \left(y+\frac{D}{4a}\right) = a\left(x+\frac{b}{2a}\right)^2$$

Let $x+\frac{b}{2a} = X$ and $y+\frac{D}{4a} = Y$

$\Rightarrow$ $Y = aX^2 \Rightarrow X^2 = \frac{Y}{a}$ hence the shape of $y = f(x)$ is a parabola

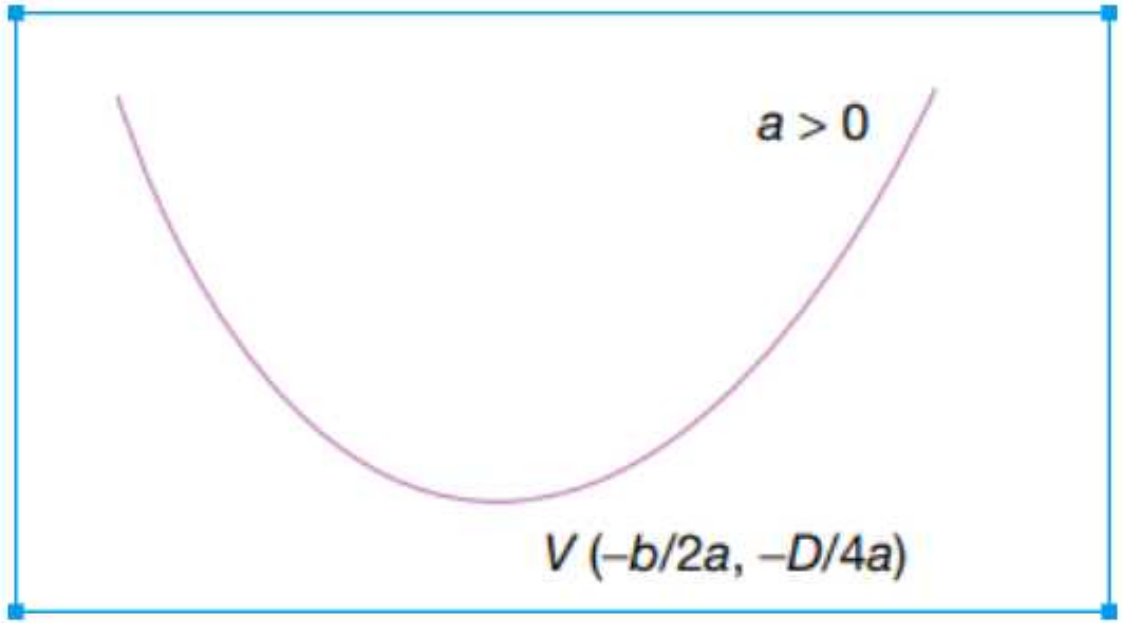
For vertex

$$X = 0 \Rightarrow x+\frac{b}{2a} = 0 \Rightarrow x = -\frac{b}{2a} \text{ and}$$

$$Y = 0 \Rightarrow y+\frac{D}{4a} = 0 \Rightarrow y = -\frac{D}{4a}$$

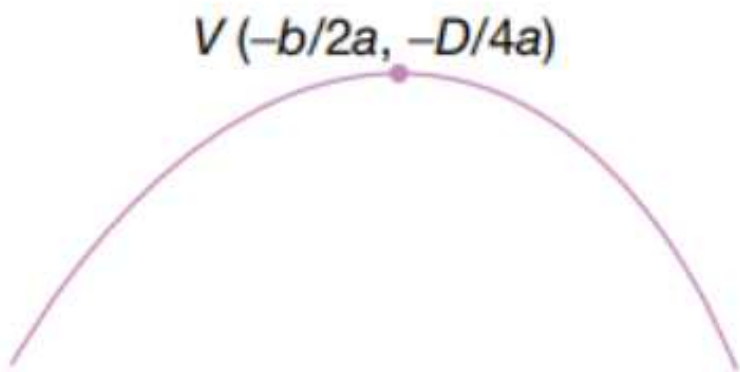
(a) If $a > 0$, the parabola opens upwards and vertex is minimum point of quadratic expression.
[least value of Quadratic Expression]

$f(x)$ is least at vertex hence when at $x = -\frac{b}{2a}$ then $y = -\frac{D}{4a}$ is minimum value of quadratic expression.



(b) If $a < 0$, then the parabola opens downwards and vertex is maximum point of quadratic expression
[Greatest value of Quadratic Expression]

$f(x)$ is greatest at vertex hence when at $x = -\frac{b}{2a}$ then $y = -\frac{D}{4a}$ is maximum value of quadratic expression.



5. Greatest and least values of $f(x)$

Vertex of the parabola $X^2 = \frac{1}{a}Y$ is

$$X = 0, Y = 0$$

$$\Rightarrow \quad x + \frac{b}{2a} = 0, y + \frac{D}{4a} = 0$$

or $$x = -\frac{b}{2a}, y = -\frac{D}{4a}$$

Hence, vertex of $y = ax^2 + bx + c$ is $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$.



For $a > 0$, $f(x)$ has least value at $x = -\frac{b}{2a}$.

This least value is given by $f\left(-\frac{b}{2a}\right) = -\frac{D}{4a}$

or $y_{\text{least}} = -\frac{D}{4a}$.

$\therefore$ Range of $y = ax^2 + bx + c$ is $\left(-\frac{D}{4a}, \infty\right)$.

For $a < 0$, $f(x)$ has greatest value at $x = -\frac{b}{2a}$.

This greatest value is given by $f\left(-\frac{b}{2a}\right) = -\frac{D}{4a}$

or $y_{\text{greatest}} = -\frac{D}{4a}$

$\therefore$ Range of $y = ax^2 + bx + c$ is $\left(-\infty, -\frac{D}{4a}\right)$.

RANGE OF QUADRATIC EXPRESSION

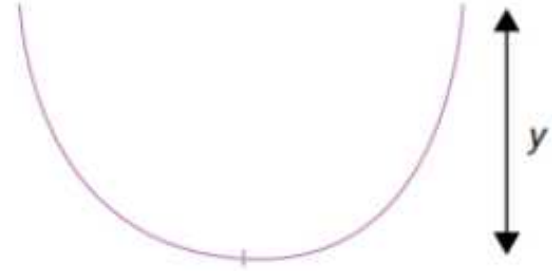
Let the quadratic expression is $y = ax^2 + bx + c$ $(a \neq 0)$

As x varies then set of values of y is known as range

Range of Quadratic Expression for all Real Values of x

CASE-1 for $a > 0$

If all real x, complete range of quadratic expression as we can see from graph is $y \in \left[-\frac{D}{4a}, \infty\right)$

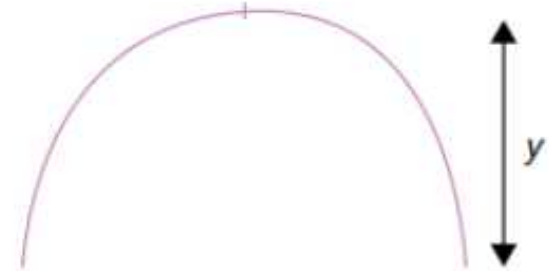


$\left(\frac{-b}{2a}, -\frac{D}{4a}\right)$

CASE-2 for $a < 0$

If all real x, complete range of quadratic expression as we can see from graph is $y \in \left(-\infty, -\frac{D}{4a}\right]$

$\left(\frac{-b}{2a}, -\frac{D}{4a}\right)$



Range of Quadratic Expression in Bounded Interval of x

A quadratic expression $y = f(x) = ax^2 + bx + c$ where $x \in [x_1, x_2]$

x-coordinate of vertex $x_0 = -\frac{b}{2a}$

Range of y if $x \in [x_1, x_2]$

$x_0 = \frac{-b}{2a} \in [x_1, x_2]$

$$\text{Range} = \left[\text{min of} \left\{\frac{-D}{4a}, f(x_1), f(x_2)\right\}, \text{max of} \left\{\frac{-D}{4a}, f(x_1), f(x_2)\right\}\right]$$

$x_0 = \frac{-b}{2a} \notin [x_1, x_2]$

$$\text{Range} = \left[\text{min of} \left\{f(x_1), f(x_2)\right\}, \text{max of} \left\{f(x_1), f(x_2)\right\}\right]$$

Sign of Quadratic Expression

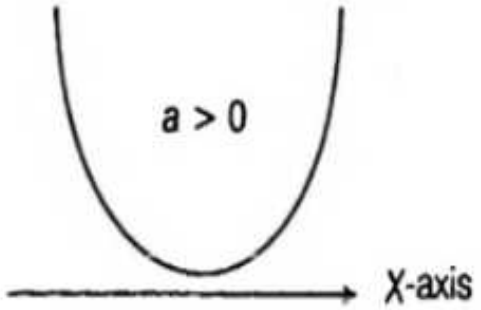
Let $f(x) = ax^2 + bx + c$ or $y = ax^2 + bx + c$,

where $a, b, c \in R$ and $a \neq 0$, for some values of x, $f(x)$ may be positive, negative or zero. This gives the following cases :

1. $a > 0$ and $D < 0$.

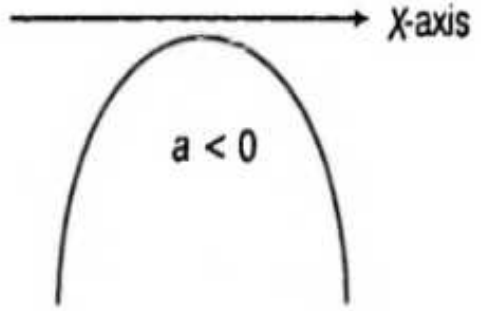
So, $f(x) > 0$ for all $x \in R$,

i.e. $f(x)$ is positive for all real values of x.



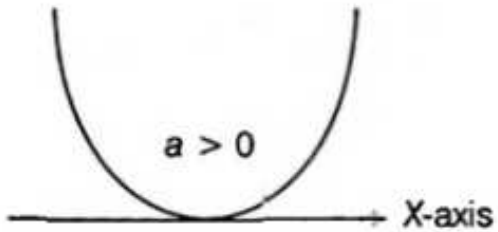
2. $a < 0$ and $D < 0$. So, $f(x) < 0$ for all $x \in R$,

i.e. $f(x)$ is negative for all real values of x.



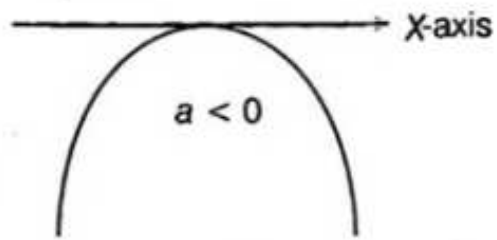
3. $a > 0$ and $D = 0$. So, $f(x) \geq 0$ for all $x \in R$,

i.e. $f(x)$ is positive for all real values of x except at vertex, where $f(x) = 0$.



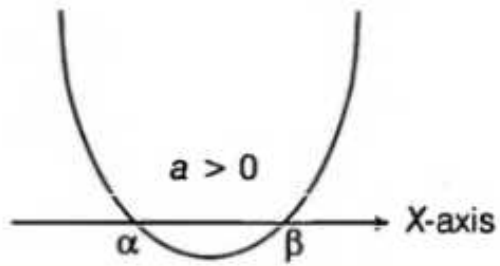
4. $a < 0$ and $D = 0$. So, $f(x) \leq 0$ for all $x \in R$,

i.e. $f(x)$ is negative for all real values of x except at vertex, where $f(x) = 0$.



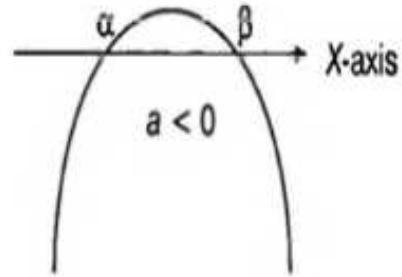
5. $a > 0$ and $D > 0$.

Let $f(x) = 0$ have two real roots α and β $(\alpha < \beta)$, then $f(x) > 0$ for all $x \in (-\infty, \alpha) \cup (\beta, \infty)$ and $f(x) < 0$ for all $x \in (\alpha, \beta)$.



6. $a < 0$ and $D > 0$

Let $f(x) = 0$ have two real roots α and β $(\alpha < \beta)$, then $f(x) < 0$ for all $x \in (-\infty, \alpha) \cup (\beta, \infty)$ and $f(x) > 0$ for all $x \in (\alpha, \beta)$.



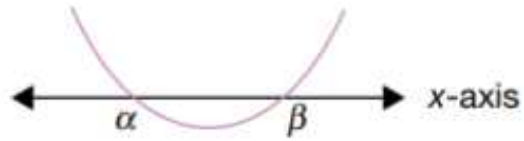
Analysis of Quadratic Expression

$y = f(x) = ax^2 + bx + c (a \neq 0)$

(a) $a > 0, D > 0$ (real and distinct)

$f(x) > 0$ for $x \in (-\infty, \alpha) \cup (\beta, \infty)$

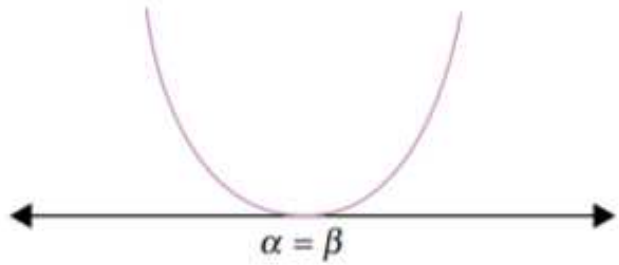
$f(x) < 0$ for $x \in (\alpha, \beta)$



(b) $a > 0, D = 0$ (real and equal roots)

$f(x) \geq 0$ for all $x \in R$

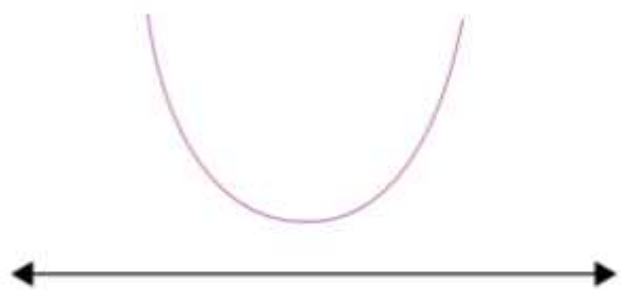
$f(x) < 0$ for $x \in \phi$



(c) $a > 0; D < 0$ (imaginary roots)

$f(x) > 0$ for $\forall x \in R$

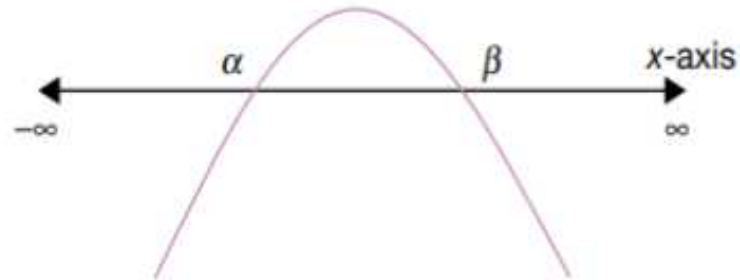
$f(x) < 0$ $x \in \phi$



(d) $a < 0; D > 0$ (real and distinct roots)

$f(x) > 0$ for $x \in (\alpha, \beta)$

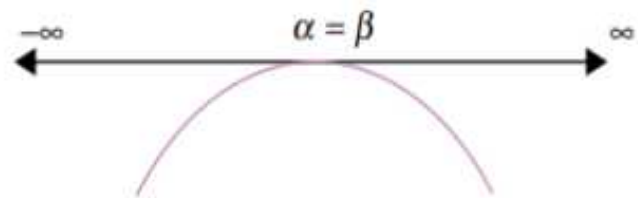
$f(x) < 0$ for $x \in (-\infty, \alpha) \cup (\beta, \infty)$



(e) $a < 0, D = 0$ (real and equal)

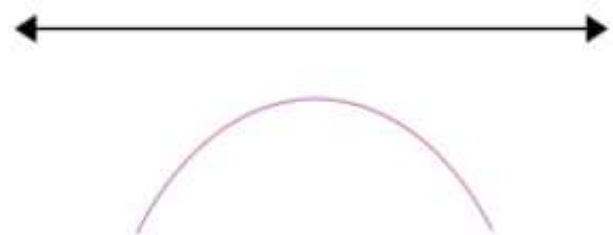
$f(x) > 0$ for $x \in \phi$

$f(x) \leq 0$ for $\forall x \in R$



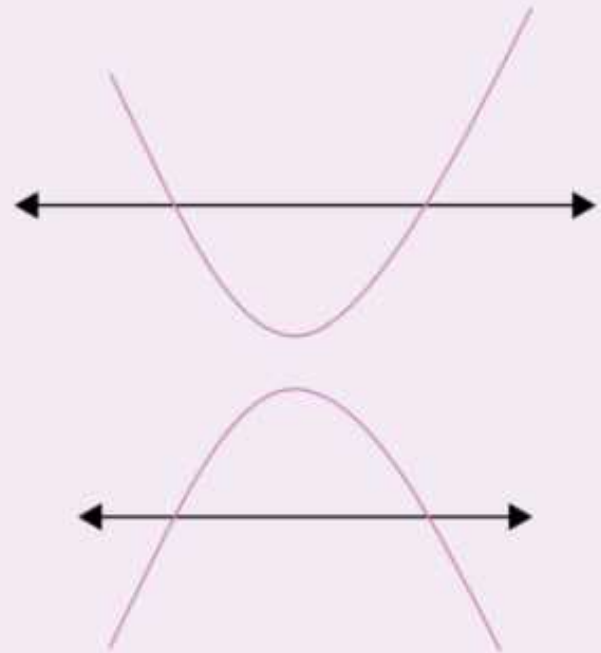
(f) $a < 0; D = 0$ (Imaginary roots)

$f(x) < 0$ for $\forall x \in R$

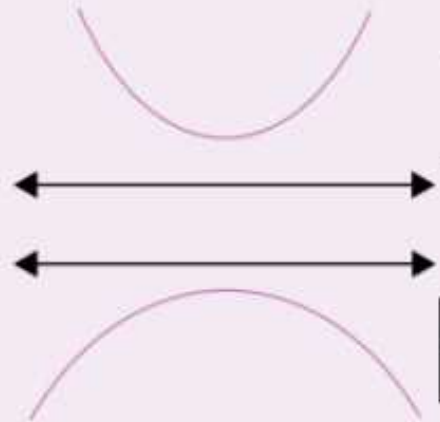


Remark(s)

(A) Quadratic expression $y = ax^2 + bx + c$ changes its sign then following two cases are possible
Condition: Discriminant $D > 0$



(B) Quadratic expression $y = ax^2 + bx + c$ does not change its sign then following cases are possible



$y = ax^2 + bx + c > 0$ for $\forall\, x \in R$
Condition $a > 0$ and $D < 0$

$y = ax^2 + bx + c > 0$ for $\forall\, x \in R$
Condition $a < 0$ and $D < 0$

Wavy Curve Method

(Generalised Method of Intervals)

Wave Curve Method is used for solving inequalities of the form

$$f(x)=\frac{(x-a_1)^{k_1}(x-a_2)^{k_2}\ldots(x-a_m)^{k_m}}{(x-b_1)^{p_1}(x-b_2)^{p_2}\ldots(x-b_n)^{p_n}}>0$$

$$(<0, \geq 0 \text{ or } \leq 0),$$

where, $k_1, k_2, \ldots, k_m, p_1, p_2, \ldots, p_n$ are natural numbers and such that $a_i \neq b_j$, where $i=1,2,\ldots,m$ and $j=1,2,\ldots,n$.

We use the following methods:

1. Solve $(x-a_1)^{k_1}(x-a_2)^{k_2}\ldots(x-a_m)^{k_m}=0$ and

   $(x-b_1)^{\rho_1}(x-a_2)^{\rho_2}\ldots(x-b_n)^{\rho_n}=0$, then we get.

   $x=a_1, a_2, \ldots, a_m, b_1, b_2, \ldots, b_n$ [critical points]

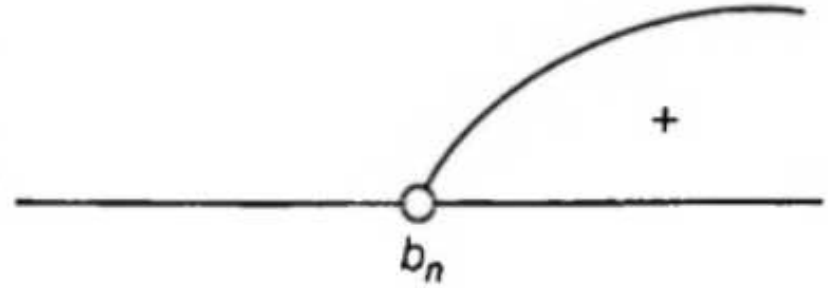
2. Assume $a_1<a_2<\ldots<a_m<b_1<b_2<\ldots<b_n$

   Plot them on the real line. Arrange inked (black) circles (●) and un-inked (white) circles (○), such that

	$a_1\ a_2\ \ldots\ a_m$	$b_1\ b_2\ \ldots\ b_n$
If $f(x)>0$	○ ○ ... ○	○ ○ ... ○
$f(x)<0$	○ ○ ... ○	○ ○ ... ○
$f(x)\geq 0$	● ● ... ●	○ ○ ... ○
$f(x)\leq 0$	● ● ... ●	○ ○ ... ○

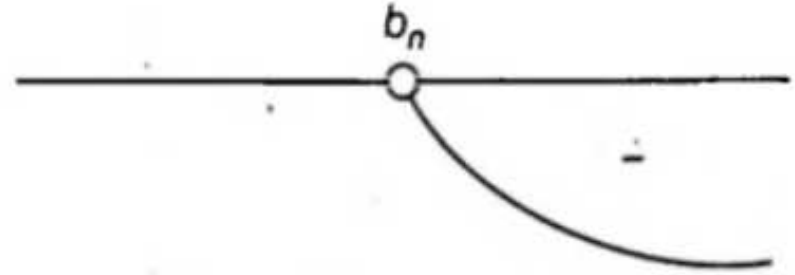
3. Obviously, b_n is the greatest root. If in all brackets before x positive sign and expression has also positive sign, then wave start from right to left, beginning above the number line, i.e.

$$+\frac{(x-a_1)^{k_1}(x-a_2)^{k_2}\ldots(x-a_m)^{k_m}}{(x-b_1)^{p_1}(x-b_2)^{p_2}\ldots(x-b_n)^{p_n}}, \text{ then}$$



and if in all brackets before x positive sign and expression has negative sign, then wave start from right to left, beginning below the number line, i.e.

$$-\frac{(x-a_1)^{k_1}(x-a_2)^{k_2}\ldots(x-a_m)^{k_m}}{(x-b_1)^{p_1}(x-b_2)^{p_2}\ldots(x-b_n)^{p_n}}, \text{ then}$$



4. If roots occur even times, then sign remain same from right to left side of the roots and if roots occur odd times, then sign will change from right to left through the roots of

$$x = a_1, a_2, \ldots, a_m, b_1, b_2, \ldots, b_n.$$

5. The solution of $f(x) > 0$ or $f(x) \geq 0$ is the union of all intervals in which we have put the plus sign and the solution of $f(x) < 0$ or $f(x) \leq 0$ is the union of all intervals in which we have put the minus sign.

Important Results

1. The point where denominator is zero or function approaches infinity, will never be included in the answer.
2. For $x^2 < a^2$ or $|x| < a \Leftrightarrow -a < x < a$

   i.e., $x \in (-a, a)$
3. For $0 < x^2 < a^2$ or $0 < |x| < a$

   $\Leftrightarrow$ $-a < x < a - \{0\}$

   i.e., $x \in (-a, a) - \{0\}$
4. For $x^2 \geq a^2$ or $|x| \geq a \Leftrightarrow x \leq -a$ or $x \geq a$

   i.e., $x \in (-\infty, -a] \cup [a, \infty)$
5. For $x^2 > a^2$ or $|x| > a \Leftrightarrow x < -a$ or $x > a$

   i.e., $x \in (-\infty, -a) \cup (a, \infty)$
6. For $a^2 \leq x^2 \leq b^2$ or $a \leq |x| \leq b$

   $\Leftrightarrow$ $a \leq x \leq b$ or $-b \leq x \leq -a$

   i.e., $x \in [-b, -a] \cup [a, b]$
7. For $a^2 < x^2 \leq b^2$ or $a < |x| \leq b$

   $\Leftrightarrow$ $a < x \leq b$ or $-b \leq x < -a$

   i.e., $x \in [-b, -a) \cup (a, b]$
8. For $a^2 \leq x^2 < b^2$ or $a \leq |x| < b$

   $\Leftrightarrow$ $a \leq x < b$ or $-b < x \leq -a$

   i.e., $x \in (-b, -a] \cup [a, b)$

9. For $a^2 < x^2 < b^2$ or $a < |x| < b$

$\Leftrightarrow$ $a < x < b$ or $-b < x < -a$

i.e., $x \in (-b, -a) \cup (a, b)$

10. For $(x - a)(x - b) < 0$ and $a < b$, then $a < x < b$

i.e., $x \in (a, b)$

11. If $(x - a)(x - b) \leq 0$ and $a < b$,

then $a \leq x \leq b$, $x \in [a, b]$

12. If $(x - a)(x - b) > 0$ and $a < b$, then $x < a$ or $x > b$

i.e., $x \in (-\infty, a) \cup (b, \infty)$

13. If $(x - a)(x - b) \geq 0$ and $a < b$,

then $x \leq a$ or $x \geq b$

i.e., $x \in (-\infty, a] \cup [b, \infty)$

